

Counting Relations

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In this document, it is assumed that each purview can be specified only once (compositional state) and N represents the system size:

Lemma 1 (From Will's proof). *The number of possible relations whose overlap is a superset of O , $|O| = K$, is*

$$F(K) = 2^{2^{N-K+1}} - (1 + 2^{N-K+1})$$

Theorem 1 (From Will's proof). *The number of possible relations is:*

$$|R| = \sum_{K=1}^N (-1)^{K-1} \binom{N}{K} F(K).$$

1 Counting Relations of a Specific Order

Theorem 2. *The number of possible relations over overlap O :*

$$N(K) = \sum_{i=0}^{N-K} (-1)^i \binom{N-K}{i} F(K+i),$$

where K is the size of O and $F(K) = 2^{2^{N-K+1}} - (1 + 2^{N-K+1})$.

$N(K)$ counts the number of relations over a specific overlap with size K . The number of all the relations over any overlap with size K is

$$\binom{N}{K} N(K)$$

and the number of all the possible relations is

$$|R| = \sum_{K=1}^N \binom{N}{K} N(K),$$

which is the same number as Theorem 1. However, this formulation gives us the ability to approximate the sum of ϕ as well or calculate an upper bound for it.

For instance, sum of ϕ_r , assuming all the relations exists and ϕ only depends on the size of the overlap:

$$\sum_{K=1}^N \binom{N}{K} N(K) \phi(K)$$

2 Counting Relations Containing a Specific Purview

Lemma 2. *The number of possible relations that contain an specific purview Z and the overlap is a superset of O ($O \subseteq Z$):*

$$F_Z(|O|) = 2^{2^{N-|O|+1}-1} - 1$$

Theorem 3. *The number of possible relations that contain an specific purview Z :*

$$\sum_{i=1}^{|Z|} (-1)^{i-1} \binom{|Z|}{i} F_Z(i)$$

Using this theorem, we can calculate what ratio of relations are killed if we remove purview Z from CES. The ratio of relations involving Z divided by all the relations:

$$\begin{aligned} & \lim_{N \rightarrow \infty} \frac{\sum_{|O|=1}^{|Z|} (-1)^{|O|-1} \binom{|Z|}{|O|} (2^{2^{N-|O|+1}-1} - 1)}{|R|} \\ &= \lim_{N \rightarrow \infty} \frac{\sum_{|O|=1}^{|Z|} (-1)^{|O|-1} \binom{|Z|}{|O|} (2^{2^{N-|O|+1}-1} - 1)}{\sum_{|O|=1}^N (-1)^{|O|-1} \binom{N}{|O|} (2^{2^{N-|O|+1}-1} - 1)} \\ &= \lim_{N \rightarrow \infty} \frac{\binom{|Z|}{1} (2^{2^N-1} - 1)}{\binom{N}{1} (2^{2^N} - (1 + 2^N))} \\ &= \lim_{N \rightarrow \infty} \frac{|Z| (2^{2^N-1})}{N (2^{2^N})} = \frac{1}{2} \frac{|Z|}{N} \end{aligned}$$

So if we remove a single purview of size N , about half of the relations are going to be killed.

Theorem 4. *The number of possible relations that contain an specific purview Z and the overlap is exactly O ($O \subseteq Z$):*

$$N_Z(|O|) = \sum_{i=0}^{|Z|-|O|} (-1)^i \binom{|Z|-|O|}{i} F_Z(|O|+i)$$

$N_Z(|O|)$ counts the number of relations that contain purview Z over a specific overlap with size $|O|$. The number of the relations that contain purview Z over

any overlap with size $|O|$ is

$$\binom{|Z|}{|O|} N_Z(|O|)$$

and the number of all the relations that contain purview Z is

$$\sum_{|O|=1}^{|Z|} \binom{|Z|}{|O|} N_Z(|O|),$$

which is the same number as Theorem 3, but more helpful if we want to calculate the sum of ϕ of such relations.